



# Modelling Financial Markets

## Chapter 13: Machine Learning in Finance



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## Acronyms used in this material (1/3)

- ▣ ADF: Augmented Dickey–Fuller (test)
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AUC: Area Under the (ROC) Curve
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ CAPM: Capital Asset Pricing Model
- ▣ CI: Confidence Interval
- ▣ CMA: Conservative Minus Aggressive (investment factor)
- ▣ CPCV: Combinatorial Purged Cross-Validation
- ▣ CRSP: Center for Research in Security Prices
- ▣ CSCV: Combinatorially Symmetric Cross-Validation
- ▣ CV: Cross-Validation
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DML: Double/Debiased Machine Learning
- ▣ DSR: Deflated Sharpe Ratio
- ▣ ELW: Exact Local Whittle (estimator)
- ▣ ES: Expected Shortfall
- ▣ EU: European Union
- ▣ EV: Explained Variation
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FDR: False Discovery Rate
- ▣ FF3: Fama–French three-factor model
- ▣ FF5: Fama–French five-factor model
- ▣ FFD: Fixed-width window Fractional Differentiation

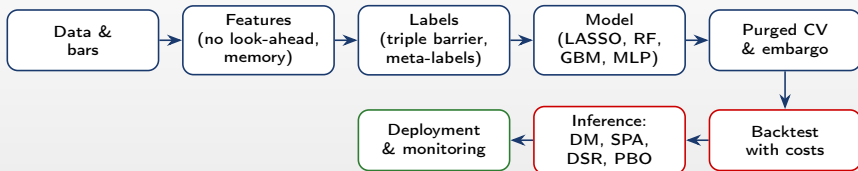
## Acronyms used in this material (2/3)

- ▣ **FFN: FeedForward Network**
- ▣ **FN: False Negative**
- ▣ **FRED-MD: Federal Reserve Economic Data, Monthly Database (McCracken–Ng)**
- ▣ **FWER: Family-Wise Error Rate**
- ▣ **GAN: Generative Adversarial Network**
- ▣ **GBM: Gradient Boosting Machine**
- ▣ **GMM: Generalized Method of Moments**
- ▣ **GRS: Gibbons–Ross–Shanken (test)**
- ▣ **GRU: Gated Recurrent Unit**
- ▣ **GW: Giacomini–White (test)**
- ▣ **HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)**
- ▣ **HML: High Minus Low (value factor)**
- ▣ **IBM: International Business Machines (Corporation)**
- ▣ **IS: In-Sample**
- ▣ **LASSO: Least Absolute Shrinkage and Selection Operator**
- ▣ **LIME: Local Interpretable Model-agnostic Explanations**
- ▣ **LLM: Large Language Model**
- ▣ **LS: Linear SDF estimated by unregularised Least Squares**
- ▣ **LSTM: Long Short-Term Memory (network)**
- ▣ **LW: Local Whittle (estimator)**
- ▣ **MA: Moving Average**
- ▣ **MA20: 20-day Moving Average**
- ▣ **MDA: Mean Decrease Accuracy**
- ▣ **MDI: Mean Decrease Impurity**
- ▣ **ML: Machine Learning**
- ▣ **MLP: MultiLayer Perceptron**

## Acronyms used in this material (3/3)

- ▣ MSCA: Marie Skłodowska-Curie Actions
- ▣ OLS: Ordinary Least Squares
- ▣ OOS: Out-Of-Sample
- ▣ PAC: Probably Approximately Correct (learning)
- ▣ PBO: Probability of Backtest Overfitting
- ▣ PCR: Principal Component Regression
- ▣ PLS: Partial Least Squares
- ▣ PSR: Probabilistic Sharpe Ratio
- ▣ PT: Pesaran–Timmermann (test of directional accuracy)
- ▣ RC: Reality Check (White's test for data snooping)
- ▣ RF: Random Forest
- ▣ RL: Reinforcement Learning
- ▣ RMW: Robust Minus Weak (profitability factor)
- ▣ ROC: Receiver Operating Characteristic (curve)
- ▣ RSI: Relative Strength Index
- ▣ SD: Standard Deviation
- ▣ SDF: Stochastic Discount Factor
- ▣ SE: Standard Error
- ▣ SHAP: SHapley Additive exPlanations
- ▣ SPA: Superior Predictive Ability (test)
- ▣ SR: Sharpe Ratio
- ▣ TN: True Negative
- ▣ TP: True Positive
- ▣ US: United States
- ▣ VIX: CBOE Volatility IndeX
- ▣ WF: Walk-Forward (validation)

## Chapter Route: the Financial ML Workflow



- ▣ Part I: the prediction problem, a minimal toolbox, features and fractional differencing, labels, honest validation
- ▣ Part II: an S&P 500 case study with formal tests, selection and backtest overfitting, machine learning (ML) for inference, a research case study, AI for scientific discovery; technical detail sits in the [Appendix](#)
- ▣ After this chapter you can: build features and labels without look-ahead; validate with purging and embargo; test a classifier and a backtest against a benchmark (PT, DM, DSR, SPA, PBO)
- ▣ **Look-ahead bias**: using information not available at decision time; **selection bias**: reporting the best of many trials; [López de Prado \(2018\)](#): most failures come from the process, not the algorithm

## Why Machine Learning in Finance, and Why It Is Hard

### The promise

- Flexible, non-linear functional forms; interactions found automatically
- Hundreds of candidate predictors (the “factor zoo”, [Chapter 3](#))
- Alternative data: text, images, order books, on-chain data

### The constraints

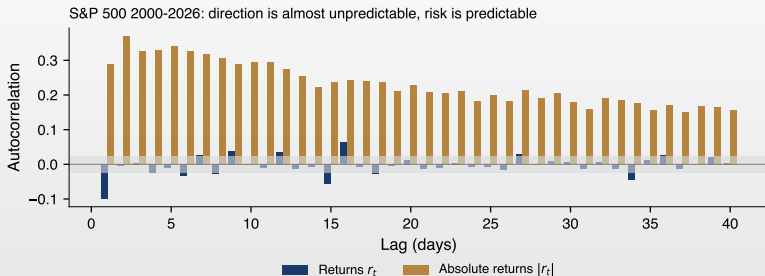
- Very low **signal-to-noise ratio**; one serially dependent history that cannot be re-run
- **Non-stationarity**: regimes, breaks; markets react to their users
- Labels must be constructed; errors cost real money
- Massive risk of **overfitting** (fitting noise)

### Adaptive Markets Hypothesis: Lo (2004)

- Predictability appears, is arbitrated away and re-appears: a model that worked yesterday is a **hypothesis**, not a law (market efficiency: [Chapter 2](#))
- The hard part is not fitting a model but **not fooling yourself**: correct labels, correct validation, honest backtests



## What Is Predictable? Direction vs. Risk



- Autocorrelation at lag  $k$ :  $\rho_k = \text{Corr}(x_t, x_{t-k})$ ; the grey band is the 95% no-correlation band  $\pm 1.96/\sqrt{T}$
- Autocorrelations of returns  $r_t$  are tiny (inside the band); those of  $|r_t|$  are large and persistent
- ML has the best chance where there is structure: **volatility, risk, liquidity, cross-sectional ranking**

## Evidence, and the Economic Value of a Small $R_{OOS}^2$

- Gu, Kelly & Xiu (2020): monthly returns of individual US stocks; linear models, penalised regressions, principal component regression (PCR), partial least squares (PLS), forests, boosted trees, neural nets
  - ▶  $R_{OOS}^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$  is small (well below 1% a month); trees and nets win through **non-linear interactions**; momentum, liquidity and volatility dominate
- Campbell & Thompson (2008): in  $r_{t+1} = \mu + x_t + \varepsilon_{t+1}$  with  $R^2 = \text{Var}(x_t) / \text{Var}(r_{t+1})$ , an unconstrained mean-variance investor who knows  $x_t$  (no costs, no estimation error) reaches

$$SR^{*2} = \frac{SR_0^2 + R^2}{1 - R^2}, \quad SR_0 = \text{Sharpe ratio of buy-and-hold}$$

- ▶ S&P 500, monthly log returns 2001–2026:  $SR_0 = 0.128$  a month (0.44 annualised);  $R^2 = 0.5\%$  a month  $\Rightarrow SR^* = 0.51$  (+15%),  $R^2 = 1\% \Rightarrow 0.57$  (+27%)
- The benchmark matters: Welch & Goyal (2008) show that most published predictors lose to the **historical mean**; Gu, Kelly & Xiu (2020) and Chapter 14 use a zero forecast
- For a **single index** the effective sample is far smaller than for thousands of stocks: the S&P 500 case study tests exactly this

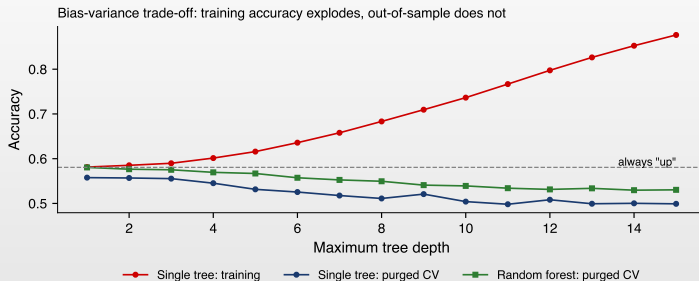
## Loss, Penalty and Five Model Families

- Supervised learning:  $\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_i L(y_i, f(x_i)) + \lambda \Omega(f)$ ; loss  $L$  (squared error, log-loss), complexity penalty  $\Omega$ , tuning  $\lambda \geq 0$  chosen by  $K$ -fold cross-validation (CV)
  - expected error = bias<sup>2</sup> + variance + noise  $\sigma^2$  (Hastie, Tibshirani & Friedman (2009), Ch. 7); in finance  $\sigma^2$  dominates, so variance control matters more than flexibility
- Penalised logit / least squares:  $\lambda [\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2]$ : Ridge  $\alpha = 0$  (Hoerl & Kennard (1970)), LASSO  $\alpha = 1$  (Tibshirani (1996)), Elastic Net in between (Zou & Hastie (2005)); the linear benchmark is the logit  $p = 1/(1 + e^{-x^\top \beta})$
- Trees: recursive splits minimising Gini impurity; random forest (RF): average of  $B$  decorrelated bootstrap trees (Breiman (2001)); its out-of-bag error is **optimistic** when labels overlap
- Gradient boosting machine (GBM):  $F_m = F_{m-1} + \eta h_m$ , small trees fitted to the negative gradient (Friedman (2001))
- Multilayer perceptron (MLP):  $h^{(\ell)} = g(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$ , ReLU  $g$ , early stopping; architectures in Chapter 14
- This chapter classifies the 5-day direction; every CV score is an estimate and is tested against a benchmark



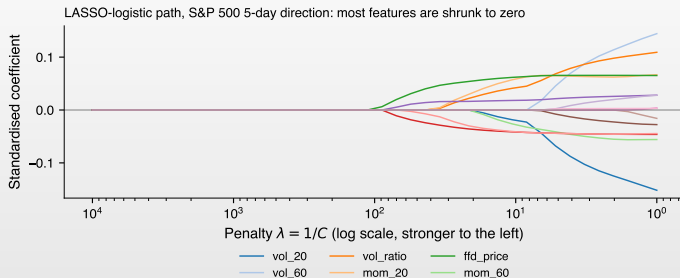
Random Forests

## Bias–Variance in Practice: Tree Depth



- S&P 500, 5-day direction; accuracy = share of correctly classified days
- Training accuracy rises to  $\approx 87\%$  with depth; purged-CV accuracy **falls** towards 50%
- Random forests degrade more slowly, but never beat the “always up” baseline (dashed line)

## LASSO Regularisation Path



- $L_1$ -penalised logistic regression for the 5-day direction of the S&P 500, 14 standardised features
- Strong penalty: every coefficient is zero; the first feature to enter is the FFD price, a scaled price level (next section)
- Weak penalty: the volatility features (20- and 60-day volatility and their ratio) get the largest coefficients
- All coefficients are small in absolute value: weak signal, consistent with near-efficient markets

## Stationarity, Memory and the Fractional Difference

- ML inputs need a distribution that does not drift; log prices behave as  $I(1)$ , returns  $\Delta \log P_t$  are  $I(0)$  but **erase the memory** of the price level
- A series is  $I(\delta)$  if  $(1 - B)^\delta X_t$  is  $I(0)$ , with backshift  $BX_t = X_{t-1}$ ; it is stationary only for  $\delta < \frac{1}{2}$  ([Hosking \(1981\)](#))

### Definition 1 (Fractional difference of order $d$ )

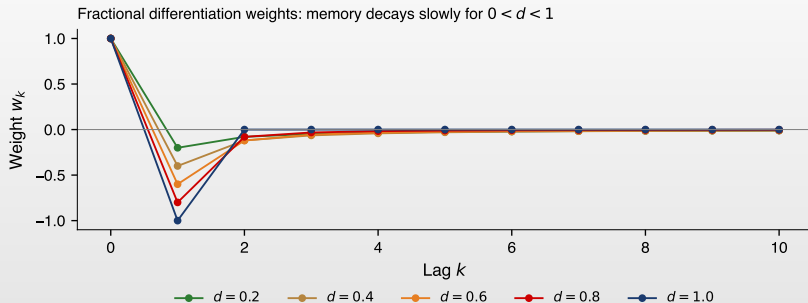
- $\tilde{X}_t = (1 - B)^d X_t = \sum_{k=0}^{\infty} w_k X_{t-k}$ , with  $w_0 = 1$  and  $w_k = -w_{k-1} \frac{d-k+1}{k}$
- $d = 1$ :  $w = (1, -1, 0, \dots)$ , ordinary returns;  $0 < d < 1$ : weights decay hyperbolically, so memory is kept
- An  $I(1)$  log price differenced with  $d$  is  $I(1 - d)$ : non-stationary for  $d \leq \frac{1}{2}$ , stationary with long memory for  $\frac{1}{2} < d < 1$
- The infinite weights sum to  $(1 - 1)^d = 0$ : the level cancels only in the limit (asymptotics in the [Appendix](#))
- [López de Prado \(2018\)](#) proposes the minimum  $d$  at which an ADF test rejects; this section shows why that rule fails

## The Fixed-Width Window (FFD)

### Definition 2 (Fixed-width-window fractional differentiation (FFD))

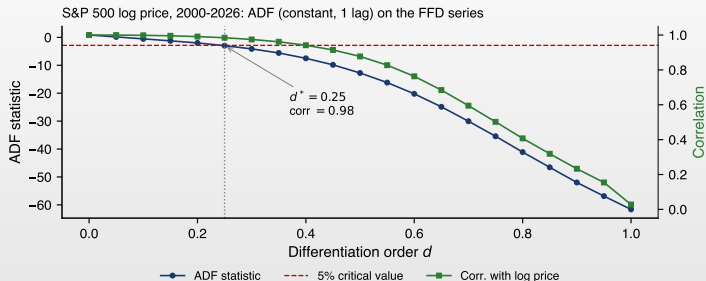
- ▣ Truncate the weights: keep  $w_0, \dots, w_K$  with  $|w_k| \geq \tau$  (e.g.  $\tau = 10^{-4}$ ), drop the rest (López de Prado (2018))
- ▣  $\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k}$  for every  $t > K$
- ▣ Every  $\tilde{X}_t$  uses the same number of lags; only past values enter  $\Rightarrow$  no look-ahead
- ▣ The smaller  $d$ , the more lags are needed:  $d = 0.25, \tau = 10^{-4} \Rightarrow K = 444$  lags
- ▣ The truncated weights no longer sum to zero:  $\sum_{k=0}^K w_k = 0.178$  at  $d = 0.25$  (0.040 at  $d = 0.5$ )

## Fractional Differentiation Weights



- For  $d = 1$  only the first lag matters; for small  $d$  the weights of distant lags remain non-zero
- All weights after  $w_0$  are negative:  $\tilde{X}_t$  compares today's level with a weighted history

## The ADF Statistic Along $d$ : S&P 500



- Augmented Dickey–Fuller (ADF) unit-root test (Time Series course):  $H_0$  unit root, reject below  $\approx -2.86$  at 5%
- FFD log price, S&P 500 2000–2026, constant and 1 lag: the ADF rejects from  $d = 0.25$ , where  $\text{Corr}(\tilde{X}_t, \log P_t) = 0.98$
- Same series, lag length chosen by AIC (34 lags): ADF =  $-0.72$ , no rejection
- A correlation of 0.98 with an  $I(1)$  price is a warning, not a virtue

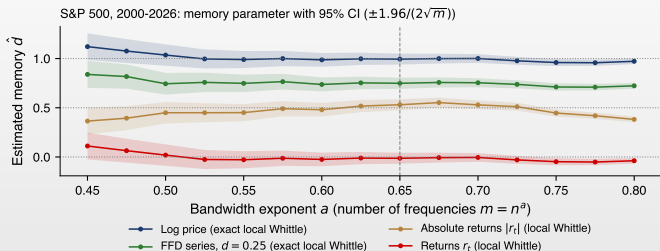
## Truncated FFD of an $I(1)$ Price Is Still $I(1)$

- Decompose the FFD series:

$$\tilde{X}_t = \sum_{k=0}^K w_k X_{t-k} = \left( \sum_{k=0}^K w_k \right) X_t - \sum_{k=1}^K w_k (X_t - X_{t-k})$$

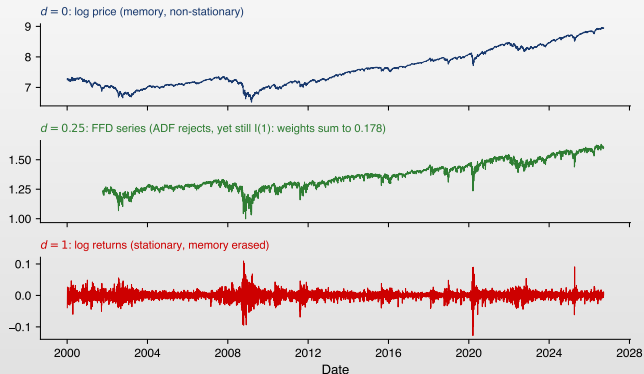
- If  $X_t$  is  $I(1)$ , each  $X_t - X_{t-k}$  is  $I(0)$ ; the first term is  $0.178 X_t$  at  $d = 0.25$ : a **scaled random walk**
- The unit-root component has small variance next to the stationary part, so ADF with few lags confuses the two
  - ▶ Monte Carlo: pure random walk,  $T = 6,274$ , FFD with  $d = 0.25$ : ADF (1 lag) rejects in **97.8%** of 500 samples, at a nominal 5%
- Consequence: the FFD log-price in the case study is a scaled price level, not a stationary feature; the next slide measures the memory directly

## Memory Estimated Directly: Local Whittle across Bandwidths



- An  $I(\delta)$  series has spectral density  $f(\lambda) \sim G\lambda^{-2\delta}$  near 0; local Whittle (LW, Robinson (1995)) fits it on the first  $m = n^\alpha$  Fourier frequencies, exact local Whittle (ELW, Shimotsu & Phillips (2005)) also for  $\delta \geq \frac{1}{2}$ ; 95% CI  $\hat{\delta} \pm 1.96/(2\sqrt{m})$
- S&P 500, 2000–2026,  $m = n^{0.65}$ : log price (ELW) 0.99 [0.94; 1.05]; FFD series (ELW) 0.75 [0.69; 0.81]; returns  $-0.01$   $[-0.07; 0.04]$ ;  $|r_t|$  0.53 [0.47; 0.59]
- The FFD series stays near  $1 - d$  over the band (its order is still 1); the long memory is in **volatility** (Chapters 5 and 9): fractional features belong on  $|r_t|$  (Franke, Härdle & Hafner (2019))

## Prices, FFD Series and Returns



- **Top:** log price —  $I(1)$
- **Middle:** FFD with  $d = 0.25$  — the upward trend of the price is still visible:  $0.178 \log P_t$  plus a stationary part
- **Bottom:** log returns — only volatility clusters are visible
- The FFD series enters the case study as the feature “FFD log-price”

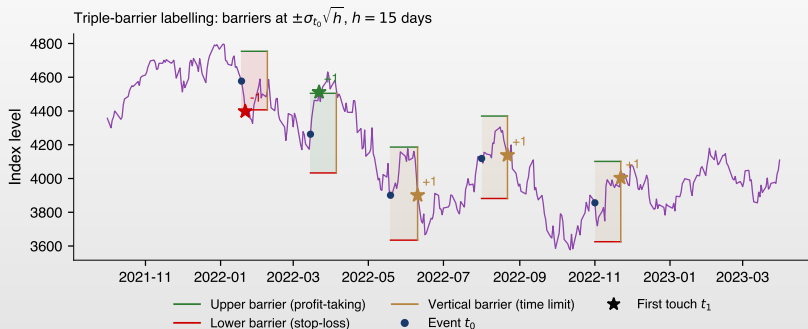
## From Fixed-Horizon to Triple-Barrier Labels

- ▣ **Fixed horizon:**  $r_{t,t+h} = \log P_{t+h} - \log P_t$ , threshold  $\tau \geq 0$ :  $y_t = +1$  if  $r_{t,t+h} > \tau$ ,  $-1$  if  $r_{t,t+h} < -\tau$ ,  $0$  otherwise
  - ▶ a constant  $\tau$  ignores time-varying volatility, the path ignores stop-losses, and the classes are unbalanced: S&P 500,  $h = 10$ ,  $\tau = 0$ : 60.5% up vs. 39.5% down

### Definition 3 (Triple-barrier label, López de Prado (2018))

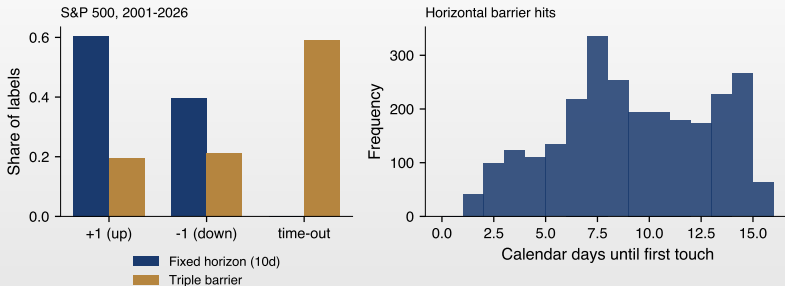
- ▣ At event time  $t_0$ , with daily volatility  $\sigma_{t_0}$  and horizon  $h$ : upper barrier  $\log P_t - \log P_{t_0} \geq \pi_u \sigma_{t_0} \sqrt{h}$  (profit-taking), lower barrier  $\leq -\pi_l \sigma_{t_0} \sqrt{h}$  (stop-loss), vertical barrier  $t = t_0 + h$
- ▣ Label = **first barrier touched** at  $t_1$ :  $+1$  upper,  $-1$  lower; at the vertical barrier  $\text{sign}(r_{t_0,t_1})$  (or  $0$ )
- ▣  $\sigma_{t_0}$ : square root of the exponentially weighted moving average (EWMA) of squared deviations of daily log returns, span 50 ( $\lambda = 1 - 2/51$ ): barriers scale with volatility, like a trader's take-profit, stop-loss and maximum holding period
- ▣ Each label has a **span**  $[t_0, t_1]$ : the input of purging in validation

## Triple Barrier: S&P 500 in 2022



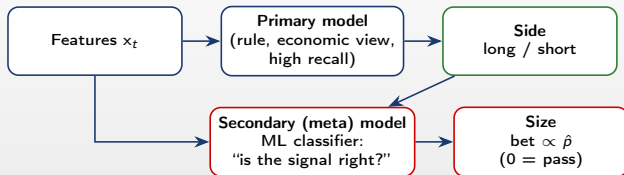
- Five events during the 2022 bear market,  $\pi_u = \pi_l = 1$ ,  $h = 15$  days
- January: stop-loss hit ( $-1$ ); March: profit-taking ( $+1$ ); the others end at the vertical barrier

## Label Distribution: Fixed Horizon vs. Triple Barrier



- S&P 500, 2001–2026,  $h = 10$ : fixed horizon gives 60.5% up / 39.5% down
- Triple barrier ( $\pm 1\sigma\sqrt{h}$ ): 19.6% up, 21.3% down, 59.0% time-out — balanced directional labels
- Right panel: horizontal barriers are typically hit after 1–2 weeks

## Meta-Labeling: Deciding Whether to Act



- Meta-label  $y_t^{meta} = 1$  if the primary signal was profitable (barrier touched in its direction), 0 otherwise; the secondary model gives  $\hat{p} = P(y_t^{meta} = 1 \mid x_t)$
- ML learns **when to trust** an interpretable primary rule, not the direction itself: precision can rise at the cost of recall
- Seminar B9, Bitcoin MA20/MA50 crossover, 1,660 test trades: precision  $0.475 \rightarrow 0.517$ , recall  $1 \rightarrow 0.541$ , F1  $0.644 \rightarrow 0.529$ ; gross return per trade  $-0.27\% \rightarrow +0.19\%$
- Filtering need not improve F1; with costs per trade, precision and the return per trade are what matter

## Overlapping Labels and Sample Uniqueness

- Label  $i$  spans  $[t_{0,i}, t_{1,i}]$ ; consecutive labels share returns  $\Rightarrow$  samples are **not i.i.d.**
- Concurrent labels at  $t$ :  $c_t = \sum_i 1\{t_{0,i} \leq t \leq t_{1,i}\}$ ; average uniqueness  $\bar{u}_i = \frac{1}{t_{1,i} - t_{0,i} + 1} \sum_{t=t_{0,i}}^{t_{1,i}} \frac{1}{c_t}$
- $\bar{u}_i = 1$ : no shared return;  $\bar{u}_i = 1/5$ : the harmonic mean of  $c_t$  over the span of label  $i$  is five

### Consequences

- Effective sample size  $\approx \sum_i \bar{u}_i \ll n$  (an overlap heuristic, not a variance-based measure)
- Bagging draws near-duplicates; CV scores too optimistic

### Remedies

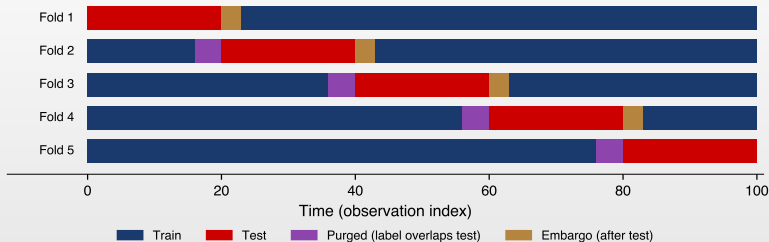
- Weight samples by  $\bar{u}_i$  (and by  $|r|$ )
- Sequential bootstrap; in RF, the fraction of observations drawn for each tree  $\approx \bar{u}$

## Leakage: What Crosses the Train/Test Boundary

### Definition 4 (Leakage)

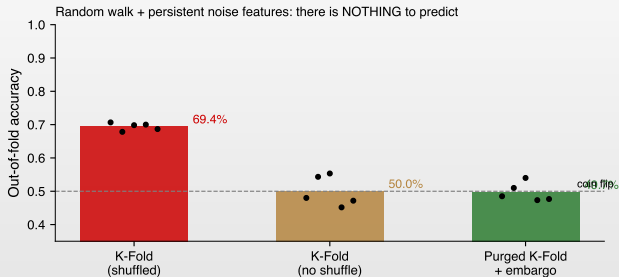
- The training set contains information that is also present in the test set, so the test score no longer measures performance on genuinely new data
- Serially correlated features ( $x_t \approx x_{t+1}$ ) and overlapping labels ( $y_t \approx y_{t+1}$ ): a shuffled  $K$ -fold puts **near-identical** days on both sides
- A flexible model (RF, GBM) memorises the neighbour  $\Rightarrow$  excellent “out-of-sample” score; tuning on a leaky CV selects the **most overfitted** model
- Other sources: transformations fitted on the full sample (scaling, principal component analysis, the choice of  $d$  for FFD), survivorship bias, data revisions, wrong timestamps

## Purged K-Fold with Embargo



- ❑ Folds are **contiguous** blocks in time (López de Prado (2018)); **purging** removes every training observation whose label span  $[t_{0,i}, t_{1,i}]$  overlaps the span of the test labels, from the first test  $t_0$  to the last test  $t_1$
- ❑ **Embargo**: also remove a small share (e.g. 1%) of observations right after the test block, because features look back in time
- ❑ One boundary: labels end 2 days after they start, test  $\{4, 5\}$ : observations 2 and 3 end inside the test block (purged), 6 is the embargo, 7 starts before the last test label ends (purged); full table in the [Appendix](#)

## Leakage Experiment: Nothing to Predict



- Price = pure random walk; label = sign of the 20-day forward return; 5 features = 20-day rolling means of independent noise
- True accuracy is 50% by construction; random forest, 5 folds
- **Shuffled K-fold: 69.4%**; contiguous K-fold 50.0%; purged K-fold + embargo 49.7%
- Shuffling fails because labels overlap *and* features persist; a result that looks too good is usually leakage

 MFM\_ch13\_purged\_cv

## Transformations on Training Data Only: the FFD Ablation

- The “minimum  $d$ ” was chosen on 2000–2026 and then used inside purged CV: the test folds helped choose a feature
  - ▶ the same ADF rule on data up to 2009 gives  $d = 0.15$ ; on the full sample  $d = 0.25$
- Case study without the FFD log-price (purged 5-fold, mean over folds)
  - ▶ accuracy: logit 57.5%, RF 56.4%, GBM 55.6%, MLP 55.5%, all below 58.1%
  - ▶ AUC: 0.483, 0.522, 0.514, 0.514
- The negative conclusion does not depend on this feature
- Correct protocol
  - ▶ estimate  $d$  (local Whittle, with its CI) inside each training fold only
  - ▶ prefer features with an economic meaning: distances to moving averages, fractional filters of  $|r_t|$

## Model Selection versus Deployment

- Ordinary  $K$ -fold is not always invalid: for a purely autoregressive model with serially uncorrelated errors it is valid (Bergmeir, Hyndman & Koo (2018))
  - ▶ it fails when the errors are serially correlated: overlapping labels, persistent features outside the model's lags; check the autocorrelation of the CV residuals first
- **Model selection:** purged  $K$ -fold, or combinatorial purged CV (CPCV), which tests on many combinations of groups and returns a **distribution** of Sharpe ratios instead of one number
- **Deployment:** walk-forward (WF), train on  $[0, T_k]$ , test on  $(T_k, T_{k+1}]$ , roll forward; realistic, but a single path with high variance
- Rule of thumb: purged  $K$ -fold or CPCV to choose, walk-forward for the final, realistic evaluation (details in the [Appendix](#))

## Case Study: Data, Target and Validation

- ▣ S&P 500 index and VIX, daily closes, 2001–2026:  $n = 6,269$  observations
- ▣ Target:  $y_t = 1\{\log P_{t+5} - \log P_t > 0\}$ , the 5-day direction; label span  $[t, t + 5]$  used for purging
- ▣ Validation: purged 5-fold CV, 1% embargo; every model gets the same folds
- ▣ Metrics: accuracy and receiver operating characteristic (ROC) area under the curve (AUC), against the “always up” baseline
  - ▶ **fold mean**: the metric computed in each of the 5 test folds, then averaged
  - ▶ **pooled**: the metric computed once on all out-of-fold predictions together; used by the tests
- ▣ Chapter data: daily S&P 500 and VIX closes of the course data

## Case Study: 14 Features and Four Models

### Features, information up to day $t$

- ▣ **Momentum:**  $\log P_t - \log P_{t-h}$ ,  $h \in \{1, 5, 20, 60, 120\}$
- ▣ **Realised volatility:** annualised SD of daily log returns over 20 and 60 days, and their ratio
- ▣ **Distance to moving average:**  $\log P_t$  minus the mean of  $\log P$  over 50 or 200 days
- ▣ **RSI(14):**  $100 - 100/(1 + RS)$ ,  $RS$  = smoothed average gain over average loss, 14 days
- ▣ **VIX:** log level and 5-day change
- ▣ **FFD** log price,  $d = 0.25$ , chosen on the full sample (ablation in the validation section)

### Models, deliberately conservative

- ▣ **Logit:** L2-penalised (Ridge), standardised features
- ▣ **RF:** 400 trees, minimum leaf 100
- ▣ **GBM:** histogram-based, depth 3, learning rate 0.03, 200 trees
- ▣ **MLP:** hidden layers of 32 and 16 units, early stopping
- ▣ Large leaves and shallow trees limit variance

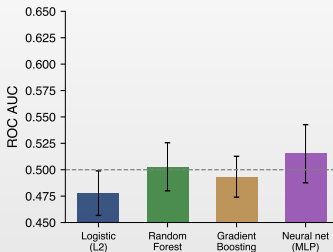
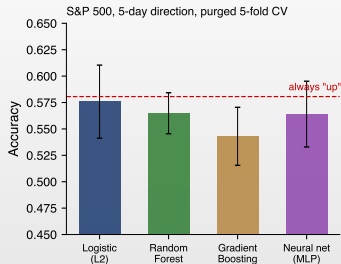
## The Model That Learned “Always Up”

|             | Pred. up   | Pred. down |
|-------------|------------|------------|
| Actual up   | TP = 3,572 | FN = 68    |
| Actual down | FP = 2,591 | TN = 38    |

- L2-penalised logit, pooled out-of-fold predictions of purged 5-fold CV,  $n = 6,269$
- TP, FN, FP, TN: true positives, false negatives, false positives, true negatives

- Accuracy =  $(TP + TN)/n = 57.6\%$ ; precision =  $TP/(TP + FP) = 0.580$ ; recall =  $TP/(TP + FN) = 0.981$ ;  $F_1 = 0.729$
- The model predicts “up” on 98.3% of days: it has essentially learned “always up”
- **Majority baseline** from the same table: actual up days  $(3,572 + 68)/6,269 = 3,640/6,269 = 58.1\%$
- A 57.6%-accurate model is *worse* than doing nothing
- Other baselines: yesterday’s direction, a moving-average rule, buy-and-hold for trading
- A model has value only if it beats the baseline **out of sample**, after costs, by more than sampling error

## Model Comparison, Fold Means



| Fold mean | Logit | RF    | GBM   | MLP   |
|-----------|-------|-------|-------|-------|
| Accuracy  | 57.6% | 56.5% | 54.3% | 56.4% |
| AUC       | 0.478 | 0.503 | 0.493 | 0.515 |

- Bars: mean over the 5 purged folds; whiskers:  $\pm 1$  SD across folds
- Baseline “always up”: **58.1%**; no model beats it

## Are the Differences Significant? Pesaran–Timmermann and AUC Intervals

- **Pesaran–Timmermann** (PT, [Pesaran & Timmermann \(1992\)](#)):  $H_0$ : predicted sign independent of realised sign;  $\hat{P}$  = share correct,  $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$ ,  $PT = (\hat{P} - \hat{P}^*) / \widehat{SE} \rightarrow N(0, 1)$  ([Appendix](#)); with overlapping labels, the slope  $t$  of  $y_t = a + b x_t + u_t$  with Newey–West errors ([Newey & West \(1987\)](#))
- **AUC**: the DeLong SE assumes independence; a circular block bootstrap ( $L = 20$  days, [Künsch \(1989\)](#)) keeps the serial dependence

| Pooled out-of-fold             | Logit          | RF             | GBM            | MLP            |
|--------------------------------|----------------|----------------|----------------|----------------|
| $\hat{P} / \hat{P}^*$          | 0.576 / 0.578  | 0.565 / 0.570  | 0.543 / 0.557  | 0.564 / 0.565  |
| $PT / HAC\ t\ (\text{lag } 4)$ | -1.28 / -1.00  | -1.61 / -1.16  | -3.07 / -2.31  | -0.36 / -0.26  |
| AUC                            | 0.479          | 0.495          | 0.490          | 0.512          |
| block-bootstrap 95% CI         | [0.455; 0.504] | [0.472; 0.520] | [0.468; 0.514] | [0.489; 0.537] |

- No significant positive directional skill; every AUC interval contains 0.5; GBM is significantly *worse* than independence
- Consistent with weak-form efficiency ([Chapter 2](#)); look at cross-sections ([Gu, Kelly & Xiu \(2020\)](#)), risk targets, crypto delisting ([Będowska-Sójka, Wójcik & Pele \(2026\)](#)), text ([Chapter 15](#))

## Comparing Forecasts: Diebold–Mariano, Clark–West, Giacomini–White

- ▣ Loss differential  $d_t = L(y_t, \hat{p}_{1,t}) - L(y_t, \hat{p}_{0,t})$ , log-loss or Brier loss  $(y_t - \hat{p}_t)^2$ ;  $d_t > 0$ : model 1 worse
- ▣ **Diebold–Mariano** (DM, Diebold & Mariano (1995)):  $DM = \bar{d} / \sqrt{\hat{\Omega}_{NW}/n} \rightarrow N(0, 1)$  under  $H_0 : \mathbb{E}[d_t] = 0$ ;  
 $\hat{\Omega}_{NW}$  = heteroskedasticity and autocorrelation consistent (HAC) long-run variance
- ▣ **Nested models** (the logit with 14 features nests the constant,  $\beta = 0$ ): under  $H_0$  the large model estimates zeros with noise,  $\mathbb{E}[d_t] > 0$ , and DM rarely favours it
- ▣ **Clark–West** (CW, Clark & West (2007)): one-sided HAC  $t$ -test of  $\mathbb{E}[f_t] = 0$ ,

$$f_t = (y_t - \hat{p}_{0,t})^2 - [(y_t - \hat{p}_{1,t})^2 - (\hat{p}_{0,t} - \hat{p}_{1,t})^2]$$

- ▶ derived for nested linear models estimated by OLS; for penalised-logit probabilities an approximation, not a theorem
- ▣ **Giacomini–White** (GW, Giacomini & White (2006)): tests the forecasting *method* as deployed, with a bounded estimation window; CW asks whether the population model adds information (statistic in the [Appendix](#))

## Feature Importance: MDI, MDA and SHAP

### Mean decrease impurity (MDI)

- Average Gini reduction from splits on a feature, over the trees
- In sample, trees only; favours continuous, high-cardinality features

### Mean decrease accuracy (MDA)

- Drop in an out-of-sample score (here AUC) when a feature is randomly permuted
- Any model; can be negative: the feature hurts

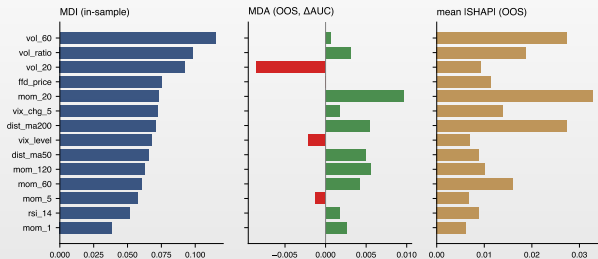
### SHAP (Lundberg & Lee (2017))

- Shapley value  $\phi_j$ : average marginal contribution of feature  $j$  over coalitions
- Additive:  $\hat{f}(x) = \phi_0 + \sum_j \phi_j$ ; mean  $|\phi_j|$  = global importance

- MDI answers “what did the model use?”; MDA and SHAP answer “what helps on new data?”
- Correlated features share and dilute importance (substitution): cluster them first; coalition formula and the conditional vs. interventional choice in the [Appendix](#)

 Shapley Values

## Feature Importance in the Case Study



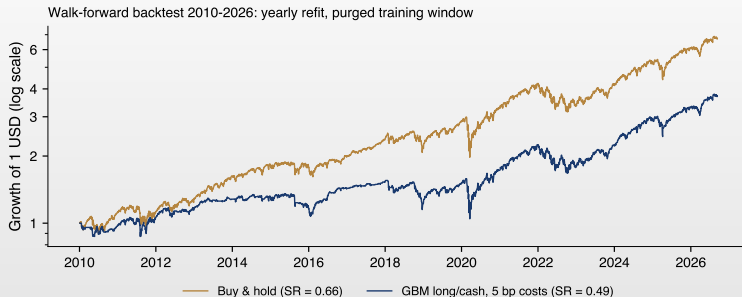
- Fit on the first 70%, evaluated on the last 30% after a purge; **RF** gives MDI and MDA, **GBM** gives SHAP: the differences are not purely methodological
- MDI ranks volatility first; MDA and SHAP point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI, **negative** MDA: a sign of overfitting, not a proof
- All importances are tiny ( $\Delta AUC < 0.01$ )
- Correlated features: permutation creates off-manifold points, so MDA and SHAP **extrapolate** (Hooker, Mentch & Zhou (2021))

 MFM\_ch13\_feature\_importance

## Walk-Forward Backtest: Strategy and Design

- ▣ Position  $w_t \in \{0, 1\}$ : long the S&P 500 if  $\hat{p}_t > 0.5$ , otherwise cash, decided with information up to  $t$
- ▣ Strategy return  $r_{t+1}^s = w_t r_{t+1} - c |w_t - w_{t-1}|$ ,  $r_{t+1}$  the daily log return;  $r - c$  is the first-order approximation of  $\log(e^r - c)$ 
  - ▶ turnover  $|w_t - w_{t-1}|$ ; cost  $c = 5$  bp (0.05%) per unit traded; exposure  $\bar{w} = \frac{1}{T} \sum_t w_t$
- ▣ Model: GBM refitted every January on all past data, purging labels that overlap the test year
- ▣ Out-of-sample period 2010–2026, one refit a year; benchmark buy-and-hold ( $w_t = 1$ : exposure 100%, no turnover)
- ▣ The only decision the backtest evaluates is the one a trader could have taken on day  $t$

## Walk-Forward Backtest: Results



- 2010–2026: strategy Sharpe **0.49** vs. buy-and-hold **0.66** (annualised); average exposure 77%; the model mostly learned “be long” and exited at the wrong times
- Paired difference  $-0.17$  (Lo SE  $\approx 0.25$  each): Ledoit–Wolf HAC SE 0.095,  $p = 0.073$  (Ledoit & Wolf (2008)); block-bootstrap CI  $[-0.36; 0.02]$ : no evidence of skill, and no significant evidence that the strategy is worse
- With  $d = 0.15$ , chosen by the same rule on data up to 2009, SR = 0.43; without the FFD log-price 0.51

## Walk-Forward: Do the Forecasts Beat a Constant?

- Walk-forward 2010–2026,  $n = 4,198$  days, 60.2% “up”; benchmark: the constant forecast = training share of “up”
- Logit (nests the constant): log-loss 0.6822 vs. 0.6757
  - ▶ DM  $t = 2.39$  ( $p = 0.017$ ): the estimated logit is significantly *worse*
  - ▶ Clark–West  $t = -0.21$  (one-sided  $p = 0.58$ ): no evidence that the features carry information
  - ▶ Giacomini–White = 5.77 ( $\chi^2_2$ ,  $p = 0.056$ ): no significant difference at 5%; approximate, since the window expands (rolling-window result in the [Appendix](#))
- GBM: log-loss 0.6971; DM  $t = 5.09$ , GW = 25.8 ( $p < 0.001$ ): worse than a constant; Pesaran–Timmermann  $-0.74$
- An honest negative result is a result: no model beats “always up” or a constant forecast

## Selection Bias and the False Strategy Theorem

- ▣ A researcher tries  $N$  strategies or configurations and reports the best; even if all have zero true Sharpe ratio,  $\max_n \widehat{SR}_n$  grows with  $N$ : **selection bias**
- ▣ Harvey, Liu & Zhu (2016): given hundreds of published factors, a new factor needs  $t > 3$ , not 2

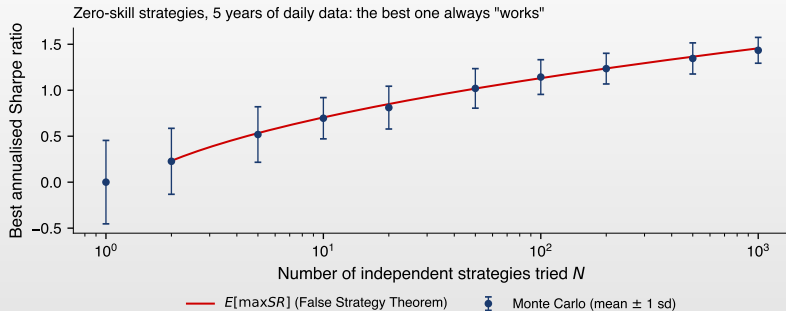
### Theorem 1 (False Strategy Theorem, Bailey & López de Prado (2014))

- ▣ If  $\widehat{SR}_n$ ,  $n = 1, \dots, N$ , are independent and approximately  $N(0, \text{Var}[\widehat{SR}])$  (no skill), the **expected maximum Sharpe ratio** is

$$SR_0 \equiv \mathbb{E} \left[ \max_n \widehat{SR}_n \right] \approx \sqrt{\text{Var}[\widehat{SR}]} \left[ (1 - \gamma) \Phi^{-1} \left( 1 - \frac{1}{N} \right) + \gamma \Phi^{-1} \left( 1 - \frac{1}{Ne} \right) \right]$$

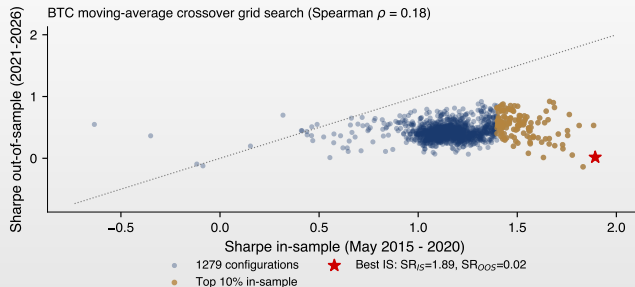
- ▣  $\gamma \approx 0.5772$ : Euler–Mascheroni constant;  $\Phi$ : standard Normal distribution function; i.i.d. returns:  $\text{Var}[\widehat{SR}] \approx 1/T$  per period

## The Best of $N$ Zero-Skill Strategies



- Strategies with i.i.d. zero-mean returns, 5 years of daily data (annualised  $\text{Var}[\widehat{SR}] \approx 1/5$ )
- Trying  $N = 1000$  variants yields a best Sharpe ratio of  $\approx 1.45$  — **with no skill at all**
- The Monte Carlo simulation agrees with the theorem

## In-Sample vs. Out-of-Sample: Bitcoin Grid Search



- Moving-average (MA) crossover on Bitcoin: long when the fast MA of the log price exceeds the slow MA by more than a band; 1,279 configurations (fast, slow, band); in sample 15 May 2015 – 2020, out of sample 2021–2026
- Selected rule (14/20 days):  $SR_{IS}$  **1.89** →  $SR_{OOS}$  **0.02** (annualised with  $\sqrt{365}$ ); Spearman rank correlation IS–OOS  $\rho = 0.18$
- Median OOS SR 0.55 for the top 10% in sample vs. 0.42 for all: the *family* of trend rules carries information, the *selected* optimum is noise

## Sharpe Ratio Inference: Standard Error and PSR

- Per-period Sharpe ratio (SR)  $\widehat{SR} = \bar{r}/\hat{\sigma}$  of daily log returns, risk-free rate 0; skewness  $\gamma_3 = \mathbb{E}[(r - \mu)^3]/\sigma^3$ , ordinary (not excess) kurtosis  $\gamma_4 = \mathbb{E}[(r - \mu)^4]/\sigma^4$
- i.i.d. returns, delta method (Lo (2002); Opdyke (2007)):  $\text{Var}(\widehat{SR}) \approx (1 - \gamma_3 \widehat{SR} + \frac{\gamma_4 - 1}{4} \widehat{SR}^2)/T$ ; Normal distribution:  $(1 + \frac{1}{2} \widehat{SR}^2)/T$

### Definition 5 (Probabilistic Sharpe Ratio (PSR), Bailey & López de Prado (2012))

- Probability that the true SR exceeds a benchmark  $SR^*$ , the variance above in the denominator:

$$\widehat{PSR}(SR^*) = \Phi \left( \frac{(\widehat{SR} - SR^*)\sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right)$$

- Annualising with  $\sqrt{q}$  needs zero autocorrelation; otherwise  $SR_q = \eta(q) SR$ ,  $\eta(q) = q/\sqrt{q + 2 \sum_{k=1}^{q-1} (q-k)\rho_k}$  (Lo (2002))
- Two strategies on the same days: test  $SR_a - SR_b$  with the HAC delta method or a studentised block bootstrap (Ledoit & Wolf (2008)), not with two separate intervals

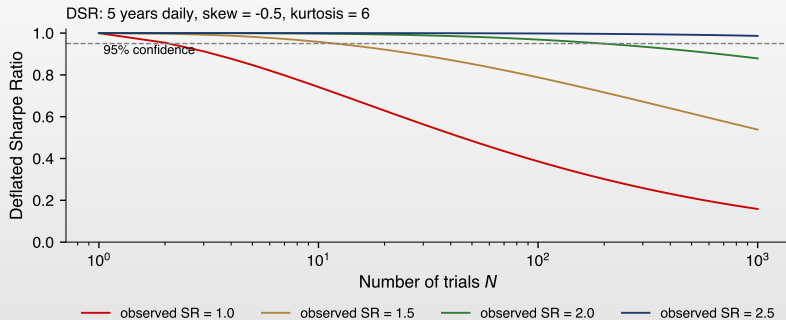
## Deflated Sharpe Ratio: the Best Bitcoin Rule

- ▣ **Deflated Sharpe Ratio** (DSR, Bailey & López de Prado (2014)):  $DSR = \widehat{PSR}(SR_0)$ , with  $SR_0$  the expected maximum of the False Strategy Theorem computed with  $\text{Var}[\widehat{SR}_n]$  of the trials
  - ▶ corrects for selection ( $N$  trials), non-Normality and sample length; reject unless  $DSR > 0.95$ , and report  $N$
- ▣ Worked example, best in-sample rule:  $T = 2,058$  days,  $\widehat{SR} = 0.0991$  a day (1.89 annualised),  $\hat{\gamma}_3 = 0.12$ ,  $\hat{\gamma}_4 = 12.33$ 
  - ▶ denominator  $\sqrt{1 - 0.12 \cdot 0.0991 + \frac{12.33-1}{4} \cdot 0.0991^2} = 1.008$ ;  $N = 1,279$  trials

| $\text{Var}[\widehat{SR}_n]$ from | SD     | $SR_0$ (per day) | $z$ / DSR    |
|-----------------------------------|--------|------------------|--------------|
| dispersion of the 1,279 trial SRs | 0.0106 | 0.035            | 2.88 / 0.998 |
| independent trials, $1/T$         | 0.0220 | 0.073            | 1.16 / 0.877 |

- ▣ The trials are highly correlated, so their dispersion understates the search: the verdict depends on this choice, and the out-of-sample SR of 0.02 sides with the conservative value

## Deflated Sharpe Ratio in Practice



- 5 years of daily returns, skewness  $-0.5$ , kurtosis 6
- An annualised SR of 1.0 is significant only if (almost) no other strategy was tried: it fails after 2–3 trials
- $SR = 1.5$  survives  $\approx 10$  trials,  $SR = 2.0 \approx 200$ ; only  $SR = 2.5$  survives 1000

## Data-Snooping Tests on the Bitcoin Grid

- $K$  rules, differentials  $d_{k,t} = r_{k,t} - r_{0,t}$  against a benchmark  $r_0$ ;  $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$ ; the null distribution comes from a **stationary bootstrap** (Politis & Romano (1994)) of whole days, which keeps the correlation between rules: no “effective”  $N$  needed
  - ▶ Reality Check (RC, White (2000)):  $\max_k \sqrt{n} \bar{d}_k$ ; Superior Predictive Ability (SPA, Hansen (2005)): studentised, clearly bad rules not recentred; Romano–Wolf StepM (Romano & Wolf (2005)): names *which* rules win, with family-wise error rate (FWER) control
- 1,279 rules, in-sample  $n = 2,058$  days, mean block 20 days, 2,000 replications (implementation in the [Appendix](#))

| Benchmark          | largest studentised mean | RC $p$ | SPA $p$ | StepM rules kept |
|--------------------|--------------------------|--------|---------|------------------|
| cash ( $r_0 = 0$ ) | –                        | 0.004  | 0.002   | 314              |
| buy-and-hold       | 0.88 (the 14/20 rule)    | 0.74   | 0.70    | 0                |

- Against cash, some trend rule earns a positive mean: Bitcoin’s own trend; no rule beats holding Bitcoin, even in sample, consistent with the OOS SR of 0.02
- DSR asks “above the luck of  $N$  trials?” and needs  $N$  and  $\text{Var}[\widehat{SR}]$  (0.998 or 0.877); SPA takes both from the data

## Probability of Backtest Overfitting (PBO)

Combinatorially symmetric cross-validation (CSCV): Bailey, Borwein, López de Prado & Zhu (2017)

1. Matrix of returns:  $T$  periods  $\times$   $N$  configurations; split the rows into  $S$  contiguous blocks
  2. For each of the  $\binom{S}{S/2}$  choices, half of the blocks are in sample (IS), the rest out of sample (OOS)
  3. Pick the best IS configuration  $n^*$ ; rank all  $N$  configurations OOS in ascending order and set  $\bar{\omega} = \text{rank}(n^*) / (N + 1) \in (0, 1)$
  4. Logit  $\lambda = \log \frac{\bar{\omega}}{1 - \bar{\omega}}$ ;  $PBO = \Pr[\lambda \leq 0]$ : the share of splits in which the IS winner ranks at or below the OOS median
- ▣  $PBO \approx 0$ : the IS winner reliably stays above the OOS median;  $PBO \approx 0.5$  or more: selecting the IS winner is no better than picking at random
  - ▣ PBO evaluates the **research procedure**, not one strategy, and is not the probability of losing money; like DSR it needs **every** trial logged (Seminar B11)

## Selection Is Not Inference: Post-Double Selection

- ▣ The LASSO path answers “which feature enters first”, not “which feature matters”; OLS  $t$ -statistics after LASSO ignore the selection step
- ▣ A control correlated with the regressor of interest  $d_t$  but only weakly with  $y_t$  gets dropped: its effect moves into the coefficient of  $d_t$  (omitted-variable bias)
- ▣ Partially linear model:  $y_t = \theta d_t + g(x_t) + \varepsilon_t$ ,  $d_t = m(x_t) + v_t$ , with  $g$ ,  $m$  approximately sparse
- ▣ Post-double selection (Belloni, Chernozhukov & Hansen (2014)):
  1. LASSO of  $y_t$  on  $x_t \Rightarrow$  selected set  $\hat{S}_y$
  2. LASSO of  $d_t$  on  $x_t \Rightarrow$  selected set  $\hat{S}_d$
  3. OLS of  $y_t$  on  $d_t$  and  $x_{\hat{S}_y \cup \hat{S}_d}$ , robust standard error
- ▣ A control is missed only if it matters for *neither* equation, so the omitted bias is second order; valid under independent sampling and sparsity, with HAC errors an approximation for time series (penalty and conditions in the [Appendix](#))

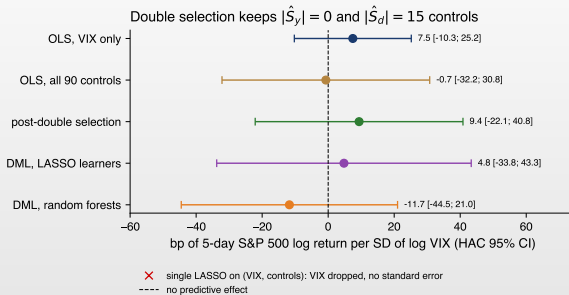
## Double/Debiased ML: Orthogonal Scores and Cross-Fitting

- Chernozhukov et al. (2018): residualise both variables with *any* learner (LASSO, RF, boosting)

$$\tilde{y}_t = y_t - \hat{\ell}(x_t), \quad \tilde{v}_t = d_t - \hat{m}(x_t), \quad \hat{\theta} = \frac{\sum_t \tilde{v}_t \tilde{y}_t}{\sum_t \tilde{v}_t^2}$$

- Neyman-orthogonal score  $\psi = (\tilde{y} - \theta \tilde{v}) \tilde{v}$ : first-order errors in  $\hat{\ell} = \hat{\mathbb{E}}[y | x]$  and  $\hat{m}$  do not bias  $\hat{\theta}$  (rate conditions in the [Appendix](#))
- Cross-fitting: fit  $\hat{\ell}$ ,  $\hat{m}$  on  $K - 1$  folds, residualise the held-out fold, repeat; removes the own-observation overfitting bias
- The published theory assumes independent sampling; with overlapping 5-day labels we use contiguous folds purged by 5 days around the test fold and HAC standard errors, a data limitation stated as such, not a new estimator

## Worked Example: Does the VIX Predict S&P 500 Returns?



- $y_t$ : 5-day forward log return (bp);  $d_t$ : log VIX, standardised (1 SD = 0.356); 90 controls: 12 technical features, squares, pairwise products;  $n = 6,269$ ; HAC errors, 4 lags
- The single LASSO drops the VIX and every control: “no effect”, with no standard error
- Double selection keeps  $|\hat{S}_y| = 0$ ,  $|\hat{S}_d| = 15$ ; 95% CI  $[-22.1; 40.8]$  bp per week per SD
- Honest intervals contain zero and large effects of both signs: the data cannot decide

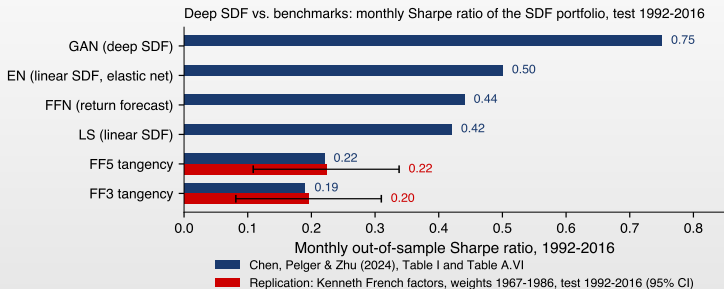
## The Virtue of Complexity, and a Critical Reading

- ▣ Kelly, Malamud & Zhou (2024): ridge regressions on  $P$  random Fourier features of a few predictors, rolling windows as short as  $T = 12$  months,  $P$  in the thousands
  - ▶ theory: with  $P > T$  the minimum-norm solution shrinks implicitly (“double descent”);  $R^2_{OOS}$  may be negative while the Sharpe ratio of market timing **rises** with complexity; complexity is not automatically overfitting
- ▣ Nagel (2025): for  $P \gg T$  the forecast is a similarity-weighted average of the  $T$  training returns; in short windows similarity is mostly recency, so the method becomes a **volatility-timed momentum** strategy, which loses on artificial data with reversals
- ▣ Avramov, Cheng & Metzker (2023): deep-learning signal profits come from hard-to-arbitrage stocks; excluding microcaps, distressed firms or high-volatility episodes attenuates them, and costs erode them further
- ▣ Chen, Pelger & Zhu (2024): impose economic structure instead; the no-arbitrage moment conditions become the loss of a deep network (next slides)
- ▣ Survey of the field: Kelly & Xiu (2023)

## Case Study: Deep Learning in Asset Pricing, Question and Method

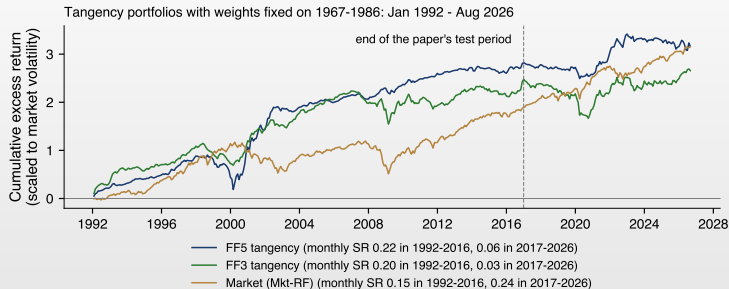
- [Chen, Pelger & Zhu \(2024\)](#): estimate the stochastic discount factor (SDF)  $M_{t+1}$  that prices **all individual US stocks**, conditioning on 46 firm characteristics and on the history of 178 macroeconomic series
  - ▶ no arbitrage:  $E_t[M_{t+1}R_{t+1,i}^e] = 0$  for every excess return; with  $M_{t+1} = 1 - \omega_t^\top R_{t+1}^e$ ,  $F_{t+1} = \omega_t^\top R_{t+1}^e$  is the conditional tangency portfolio and  $E_t[R_{t+1,i}^e] = \beta_{t,i} E_t[F_{t+1}]$  (Section I.A, eq. 1)
- Why: forecasting ([Gu, Kelly & Xiu \(2020\)](#)), the benchmark feedforward network, FFN) fits a second moment where signal-to-noise is low; the no-arbitrage condition targets a first moment, and averaging over time removes much of the noise
- Method: adversarial generalized method of moments (GMM),  $\min_{\omega} \max_g \frac{1}{N} \sum_j (E[M_{t+1}R_{t+1,j}^e g(l_t, l_{t,j})])^2$  (eq. 3, after [Hansen & Jagannathan \(1997\)](#)); an FFN for  $\omega$ , a long short-term memory network (LSTM) that turns the macro history into 4 states, an adversary  $g$  that builds the hardest test assets; the full model is labelled GAN
- Chronological split: training 1967–1986, validation 1987–1991, test 1992–2016; configurations ranked by the validation Sharpe ratio (data and tuning in the [Appendix](#))

## Deep Learning in Asset Pricing: Sharpe Ratios, Paper and Replication



- Blue: the paper, monthly out-of-sample SR 1992–2016: GAN 0.75, elastic net linear SDF (EN) 0.50, FFN 0.44, unregularised linear SDF (LS) 0.42 (Table I); FF5 0.22, FF3 0.19 (Table A.VI); the regularised no-arbitrage model already beats the forecasting network
- Red: FF3 and FF5 tangency portfolios  $w \propto \hat{\Sigma}^{-1} \hat{\mu}$  on the [Kenneth French](#) factors, weights fixed on 1967–1986: FF5 0.22 [0.11; 0.34], FF3 0.20 [0.08; 0.31] (SE 0.058, [Lo \(2002\)](#)): both published values lie inside the intervals

## Deep Learning in Asset Pricing: the Benchmarks after 2016



- Same fixed weights, January 1992 – August 2026, each series scaled to the volatility of the market excess return on 1992–2016; monthly SR 2017–2026 ( $T = 116$ ,  $SE \approx 0.09$ ): FF5 0.06, FF3 0.03, market 0.24; on 1992–2016: 0.22, 0.20, 0.15
- Replicable with public data: the Fama–French benchmarks; not replicable: the stock-level GAN, LS and EN (the CRSP panel is not in the course data); the post-2016 fall is a warning, not yet a test, and the same question applies to a GAN frozen in 1991 (details in the [Appendix](#))

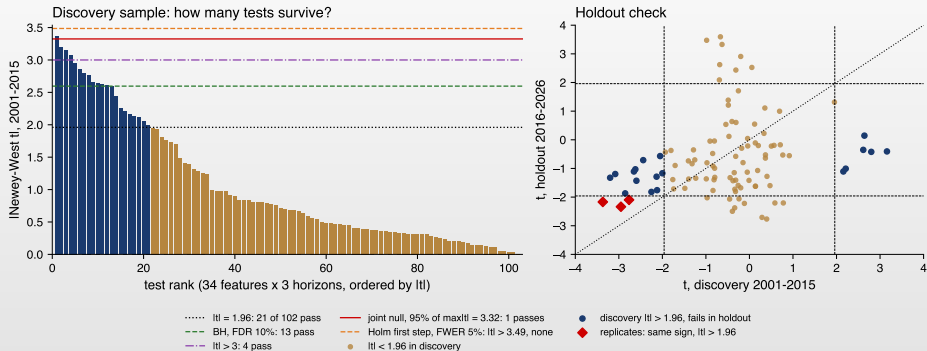
## An Open Question: Discovery or Data Mining at Scale?

- ▣ [Chen, Lopez-Lira & Zimmermann \(2022\)](#): mining 29,000 accounting ratios for  $t > 2$  gives cross-sectional predictability similar to peer-reviewed predictors; for both, about half remains after the original samples
- ▣ [Novy-Marx & Velikov \(2025\)](#): an LLM (large language model) pipeline mines over 30,000 signals, keeps 96 that pass a fixed protocol and writes hundreds of complete papers, with invented motivations and some imagined citations: HARKing (hypothesising after the results are known) at scale
- ▣ Open: when a machine proposes features, model and story, what is the number of trials, and what separates a discovery from the best of many draws?
- ▣ The discovery loop: literature  $\rightarrow$  competing hypotheses  $\rightarrow$  data and code  $\rightarrow$  robustness fixed in advance  $\rightarrow$  referee-style critique  $\rightarrow$  report every test (reusable prompts in the [Appendix](#))
- ▣ What the human checks: every reference resolves (DOI, arXiv); every trial is logged, so  $M$  in the correction counts them all; correlated tests get a joint null; transformations live inside the training fold

## Mini-Case: 102 Tests, One Old Survivor

- A typical assistant proposal: “add more indicators: momentum, moving-average distances, volatility ratios, RSI and VIX changes over several windows”
  - ▶ 34 features  $\times$  3 horizons (1, 5, 20 days) = 102 predictive regressions on 2001–2015, Newey–West  $t$ ;  $H_{0,j} : \beta_j = 0$
- $|t| > 1.96$ : 21 of 102, against 5.1 expected under the global null at 5%
- Holm (FWER 5%): 0; Benjamini–Hochberg (BH, false discovery rate (FDR) 10%, [Benjamini & Hochberg \(1995\)](#)): 13;  $|t| > 3$  ([Harvey, Liu & Zhu \(2016\)](#)): 4
- Joint null (500 circular shifts of the target): 95% quantile of  $\max_j |t_j| = 3.32$ ; one survivor, the 1-day return on the next day's return,  $t = -3.37$ : short-term reversal, a known effect
- Holdout 2016–2026: 3 of the 21 keep their sign with  $|t| > 1.96$ , all reversal variants; the VIX-based “finds” fail
- Holm and BH control different error rates at different levels; BH needs independence or positive dependence of the true-null p-values, which correlated predictors do not ensure

## Mini-Case: Thresholds and the Holdout



- Left: the 102 discovery  $|t|$  in decreasing order; each threshold answers a different question, and only the joint null accounts for the correlation between the tests
- Right: discovery vs. holdout  $t$ ; the three replications (red) are all short-term reversal; the significant VIX features collapse towards zero

## Your Turn: a Project Seed

- ▣ **Question:** can an AI-assisted feature search find a predictor of index returns (S&P 500, BET, Bitcoin) that survives a pre-registered correction and a holdout, beyond short-term reversal?
- ▣ **Replicate first:** this mini-case (21 of 102, one survivor at the joint null)
- ▣ **Deliverable:** a feature list fixed before the data are touched; all tests with Holm, BH and the max- $|t|$  null; a holdout; DSR and PBO for any strategy built on the survivors
- ▣ **Log:** each prompt that shaped a decision in AI\_USE.md; each invented reference, leak or undercounted trial in AI\_ERRORS.md, with the test that caught it ([templates](#))

## Key Takeaways and a Decision Checklist

### Three empirical lessons

- An ADF rejection is not stationarity: truncated FFD of the S&P 500 log price stays  $I(1)$ ; estimate memory with local Whittle
- Shuffled  $K$ -fold reports 69% on pure noise; on the S&P 500 no model beats “always up” (PT, DM, GW)
- The best of many backtests must be deflated or bootstrapped: Bitcoin grid 1.89 in sample, 0.02 out of sample

### Next

- Chapter 14: deep learning and time-series foundation models; Chapter 15: large language models and sentiment

### Before any claim

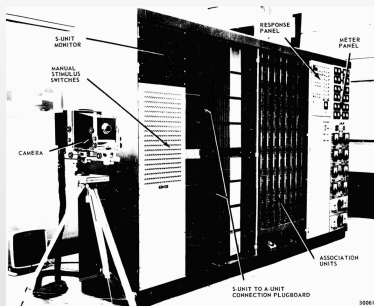
- Point-in-time data; transformations fitted on training folds only
- Volatility-scaled labels with recorded spans
- Purged  $K$ -fold or CPCV with embargo; walk-forward for the final test
- A naive baseline; PT, DM / CW / GW and block-bootstrap intervals
- Costs included; every trial logged; DSR, SPA and PBO
- “Selected” or “important” is not “significant”: double selection or DML for inference

## Appendix: Further Study on Quantinar

### Advanced courses on quantinar.com

- ☐  Random Forests
- ☐  Shapley Values
- ☐  Shapley Curves: A Smoothing Perspective
- ☐  Machine Learning in Financial Risk
- ☐  Introduction to Reinforcement Learning
- ☐  Generative Adversarial Networks
- ☐  MSCA Digital: The Need for Explainable AI
- ☐  Business Analytics and Data Science
- ☐  Model Selection Criteria
- ☐  PAC Learning

## Appendix: From the Perceptron to Deep Blue



Mark I Perceptron, operator's manual (1960)

Figure: J.C. Hay, A.E. Murray (1960); public domain; Wikimedia Commons



IBM Deep Blue, Computer History Museum

Photo: Anton Chiang (2011); CC BY 2.0; Wikimedia Commons

- **Perceptron:** a learning rule for a single artificial neuron [Rosenblatt \(1958\)](#); the Mark I was hardware built to recognise simple images
- 1997: IBM's Deep Blue beats world chess champion Garry Kasparov [Campbell, Hoane & Hsu \(2002\)](#)
- Deep Blue searched with hand-built rules; modern ML (this chapter) *learns* the rules from data

## Appendix: the In-Sample $R^2$ of a Predictor without Skill

- A monthly  $R^2$  of 0.5% is economically large, but sampling noise is of the same order
- In-sample OLS  $R^2$  of one predictor with no skill (intercept, independent errors,  $n \approx 300$  months):  $nR^2 \approx \chi_1^2$ 
  - ▶ so  $R^2$  has mean  $1/n \approx 0.3\%$  and standard deviation  $\sqrt{2}/n \approx 0.5\%$
- $R_{OOS}^2$  can be negative, has no such null distribution and needs its own test (Clark–West)

## Appendix: Worked Example: How a Forest Averages Its Trees

- Random forest of the case study (400 trees, minimum leaf size 100), trained on the first four purged folds
- Prediction for the last observation (11 September 2026):  $P(\text{up in 5 days})$  from each tree
  - ▶ individual trees range from 0.31 to 0.88 (standard deviation 0.11)
  - ▶ 81.5% of the trees vote “up” ( $p > 0.5$ )
  - ▶ forest probability = average of the 400 trees = 0.608
- Single trees disagree strongly; the average is far more stable
- Stability is not accuracy: the forest still does not beat “always up” (case study)

## Appendix: Sampling Financial Data: Bars

### Time and tick bars

- ▣ A **bar** = one row (open, high, low, close, volume) summarising a sampling interval
- ▣ Time bars: every fixed interval (1 minute, 1 day); oversample quiet periods; fat-tailed returns
- ▣ Tick bars: one bar every  $N$  transactions

### Volume, dollar and information bars

- ▣ One bar every  $V$  shares or every  $D$  dollars traded: sampling by **information arrival**
- ▣ Returns closer to i.i.d. draws from the Normal distribution: [Clark \(1973\)](#); [Ané & Geman \(2000\)](#)
- ▣ Imbalance / runs bars: sample when order flow becomes one-sided

- ▣ This chapter uses daily time bars; tick data (crypto exchanges) in [Chapters 10](#) and [16](#)

## Appendix: Fractional Weights and Their Asymptotics

### Definition 6 (Fractional difference operator)

- With the backshift operator  $BX_t = X_{t-1}$  and a real order  $d$ :

$$(1 - B)^d = \sum_{k=0}^{\infty} \binom{d}{k} (-B)^k, \quad \tilde{X}_t = \sum_{k=0}^{\infty} w_k X_{t-k}$$

- Weights:  $w_0 = 1$ ,  $w_k = -w_{k-1} \frac{d-k+1}{k} = \frac{\Gamma(k-d)}{\Gamma(-d)\Gamma(k+1)} \sim \frac{k^{-(1+d)}}{\Gamma(-d)}$  as  $k \rightarrow \infty$

- $d = 1$ :  $w = (1, -1, 0, 0, \dots)$  — ordinary returns
- $0 < d < 1$ : weights decay hyperbolically  $\Rightarrow$  long memory is kept
- Infinite sum:  $\sum_{k=0}^{\infty} w_k = (1 - 1)^d = 0$ , so the level of  $X_t$  cancels only in the limit

## Appendix: Local Whittle and Exact Local Whittle

- Periodogram  $I(\lambda_j)$  at  $\lambda_j = 2\pi j/n$ ; local Whittle ([Robinson \(1995\)](#)) on the first  $m$  frequencies:

$$\hat{\delta} = \arg \min_{\delta} \log \left( \frac{1}{m} \sum_{j=1}^m \lambda_j^{2\delta} I(\lambda_j) \right) - \frac{2\delta}{m} \sum_{j=1}^m \log \lambda_j, \quad \sqrt{m}(\hat{\delta} - \delta) \xrightarrow{d} N(0, \frac{1}{4})$$

- Exact local Whittle (ELW, [Shimotsu & Phillips \(2005\)](#); unknown mean: [Shimotsu \(2010\)](#)): periodogram of  $(1 - B)^{\delta} X_t$ , valid also for non-stationary  $\delta \geq \frac{1}{2}$
- Why the FFD series looks like  $\hat{\delta} \approx 0.75$ : above frequency  $1/K$  the truncated filter acts as  $(1 - B)^{0.25}$ ; only  $n/(2\pi K) \approx 2$  of the  $m = 294$  frequencies lie below  $1/K$ , where the unit root returns
  - the estimate is the apparent memory over the band, not the integration order of the series

 MFM\_ch13\_ffd

## Appendix: Worked Example: Purging with 10 Observations

- Observations  $t = 0, \dots, 9$ ; label of  $t$  ends at  $t + 2$  (capped at 9)
- 5 contiguous folds of 2; embargo = 1 observation

| Test | Removed    | Train      |
|------|------------|------------|
| 0, 1 | 2, 3       | 4–9        |
| 2, 3 | 0, 1, 4, 5 | 6–9        |
| 4, 5 | 2, 3, 6, 7 | 0, 1, 8, 9 |
| 6, 7 | 4, 5, 8, 9 | 0–3        |
| 8, 9 | 6, 7       | 0–5        |

- Test  $\{4, 5\}$ : labels of 2 and 3 end at 4 and 5  $\Rightarrow$  purged; 6 is the embargo; 7 starts when the last test label ends  $\Rightarrow$  purged
- With overlapping labels, up to 4 of 10 observations leave the training set

## Appendix: When Is K-Fold Valid for Time Series?

- Bergmeir, Hyndman & Koo (2018): for a **purely autoregressive** model  $y_t = g(y_{t-1}, \dots, y_{t-p}) + \varepsilon_t$  with serially uncorrelated errors, standard  $K$ -fold CV is valid
  - ▶ the errors in the test fold are uncorrelated with the training errors, so the CV error is unbiased for the OOS error
- It fails when the **errors** are serially correlated
  - ▶ overlapping labels:  $h$ -day forward returns share  $h - 1$  daily returns
  - ▶ persistent features that are not in the model's lag set (rolling means, levels such as the FFD log-price)
  - ▶ under-specified dynamics (too few lags)
- The leakage experiment isolates exactly this: 20-day labels and 20-day rolling features
- Check before choosing: the autocorrelation of the CV residuals; if it is significant, purge and embargo

## Appendix: Walk-Forward and Combinatorial Purged CV

### Walk-forward (WF)

- Train on  $[0, T_k]$ , test on  $(T_k, T_{k+1}]$ , roll forward
- Mimics real-time deployment; no future data in training
- But a **single path**: high variance, easy to overfit by repeated trials

### Combinatorial purged CV (CPCV)

- $N$  groups; test on all  $\binom{N}{k}$  combinations of  $k$  groups, with purging and embargo
- $\varphi = \frac{k}{N} \binom{N}{k}$  backtest paths; e.g.  $N = 6, k = 2$ : 15 splits, 5 paths
- A **distribution** of Sharpe ratios instead of one number

- Rule of thumb: purged K-fold / CPCV for model selection; WF for the final, realistic evaluation

## Appendix: Testing Directional Accuracy: Pesaran–Timmermann

- $H_0$ : the predicted sign  $x_t$  is independent of the realised sign  $y_t$  (Pesaran & Timmermann (1992))
- $\hat{P}$  = share of correct signs; under  $H_0$  the expected share is  $\hat{P}^* = \hat{p}_y \hat{p}_x + (1 - \hat{p}_y)(1 - \hat{p}_x)$

$$PT = \frac{\hat{P} - \hat{P}^*}{\sqrt{\hat{V}(\hat{P}) - \hat{V}(\hat{P}^*)}} \xrightarrow{d} N(0, 1), \quad \hat{V}(\hat{P}) = \frac{\hat{P}^*(1 - \hat{P}^*)}{n}$$

$$\blacktriangleright \hat{V}(\hat{P}^*) = [(2\hat{p}_y - 1)^2 \hat{p}_x(1 - \hat{p}_x) + (2\hat{p}_x - 1)^2 \hat{p}_y(1 - \hat{p}_y)]/n + 4\hat{p}_y \hat{p}_x(1 - \hat{p}_y)(1 - \hat{p}_x)/n^2$$

- Logit above:  $\hat{p}_y = 0.581$ ,  $\hat{p}_x = 0.983 \Rightarrow \hat{P}^* = 0.578 > \hat{P} = 0.576$ ;  $PT = -1.28$ , one-sided  $p = 0.90$
- PT conditions on the marginal frequencies automatically: no need to guess the baseline
- Overlapping 5-day labels: PT assumes independent days; the same null as the slope  $t$ -test in  $y_t = a + b x_t + u_t$  with Newey–West errors (Newey & West (1987)):  $t = -1.00$

## Appendix: ROC AUC and Its Sampling Uncertainty

- ▣ Receiver operating characteristic (ROC) curve: true positive rate  $TP/(TP + FN)$  against false positive rate  $FP/(FP + TN)$  over all thresholds
- ▣ Area under the curve (AUC) =  $P(\hat{p}_{\text{pos}} > \hat{p}_{\text{neg}}) + \frac{1}{2}P(\hat{p}_{\text{pos}} = \hat{p}_{\text{neg}}) = U/(n_1 n_0)$ , the Mann–Whitney statistic; 0.5 = no ranking ability
- ▣ DeLong standard error (DeLong, DeLong & Clarke-Pearson (1988)): from the placement values of each observation; assumes independent observations
- ▣ Circular block bootstrap (Künsch (1989)): resample blocks of  $L = 20$  consecutive days, 1,000 replications; keeps serial dependence
- ▣ Logit, pooled out-of-fold AUC = 0.479
  - ▶ DeLong 95% CI [0.465; 0.494]: “significantly worse than random”
  - ▶ block bootstrap 95% CI [0.455; 0.504]: noise around 0.5
- ▣ An AUC below 0.5 is anti-ranking only if it survives a test that respects the dependence

## Appendix: Giacomini–White, Statistic and Rolling Window

- **Giacomini & White (2006)**: the null concerns the forecasting **method** (model + estimation window + refitting rule), not the population model

$$H_0 : \mathbb{E}[d_{t+\tau} \mid \mathcal{F}_t] = 0 \Rightarrow \mathbb{E}[h_t d_{t+\tau}] = 0, \quad GW = n \bar{Z}^\top \hat{\Omega}^{-1} \bar{Z} \xrightarrow{d} \chi_q^2$$

- ▶  $Z_t = h_t d_{t+\tau}$ ; test functions  $h_t$  (here 1 and the lagged loss differential,  $q = 2$ );  $\tau = 5$ -day horizon
- The theory requires a bounded (rolling) estimation window, so that estimation noise never vanishes; an expanding window lies outside the theorem
- CW: does the *population* model with features add information? GW: does *my estimated* rule, as deployed, forecast better?
- S&P 500 logit, rolling 8-year window, as the GW theory requires:  $GW = 26.5$  ( $p < 0.001$ ): the logit forecasts worse than the constant



## Appendix: Feature Importance: SHAP

### Definition 7 (Shapley value of feature $j$ (SHAP, Lundberg & Lee (2017)))

- Average marginal contribution of  $j$  over all coalitions  $S \subseteq \{1, \dots, p\} \setminus \{j\}$ :

$$\phi_j = \sum_S \frac{|S|! (p - |S| - 1)!}{p!} [v(S \cup \{j\}) - v(S)], \quad \hat{f}(x) = \phi_0 + \sum_j \phi_j$$

- Interventional SHAP:  $v(S) = \mathbb{E}[\hat{f}(x_S, X_{\bar{S}})]$ , averaging  $\hat{f}$  with the features outside  $S$  drawn from a background sample (as in the Quantlet);  $\phi_0 = v(\emptyset) =$  average prediction

- Conditional SHAP uses  $v(S) = \mathbb{E}[\hat{f}(X) \mid X_S = x_S]$ ; the two differ when features are dependent
- Additive attribution of each prediction; mean  $|\phi_j|$  over observations = global importance
- **Substitution effects**: correlated features share (and dilute) importance — cluster them first



## Appendix: Reading Feature Importances

- **MDI** ranks volatility features first; **MDA** and **SHAP** point to 20-day momentum and the distance from the 200-day moving average
- 20-day volatility: high MDI but **negative** MDA — a sign of overfitting, not a proof: sampling noise and correlated features also move MDA
- Importances are tiny ( $\Delta\text{AUC} < 0.01$ ), consistent with the model comparison
- Permuting one of several correlated features creates points off the data manifold: MDA and SHAP then **extrapolate** ([Hooker, Mentch & Zhou \(2021\)](#)); permute within clusters or refit without the feature
- Importances are estimates and need their own uncertainty analysis; honest trees give asymptotically Normal forest *predictions* and CIs ([Wager & Athey \(2018\)](#)), not CIs for importances
- **Backtesting is not a research tool**, importance analysis is: [López de Prado \(2018\)](#)

## Appendix: Data-Snooping Tests: Reality Check, SPA, Romano–Wolf

- $K$  rules, performance differentials  $d_{k,t} = r_{k,t} - r_{0,t}$  against a benchmark  $r_0$ ;  $H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$
- Reality Check (RC, [White \(2000\)](#)): statistic  $\max_k \sqrt{n} \bar{d}_k$ ; null distribution from the **stationary bootstrap** ([Politis & Romano \(1994\)](#)) of the whole  $n \times K$  matrix
  - ▶ resampling whole days keeps the correlation between rules: no need to count “effective” trials
- Superior Predictive Ability (SPA, [Hansen \(2005\)](#)): studentised  $\max_k \sqrt{n} \bar{d}_k / \hat{\omega}_k$ ; clearly bad rules are not recentred, so adding junk rules does not destroy power
- Romano–Wolf StepM ([Romano & Wolf \(2005\)](#)): step-down procedure that names *which* rules beat the benchmark, with family-wise error rate (FWER) control
- DSR asks “is the best SR above the luck of  $N$  trials?” and needs  $N$  and  $\text{Var}[\widehat{SR}]$ ; these tests take both from the data

## Appendix: Double Selection and DML, Penalty and Conditions

- ▣ Partially linear model:  $\mathbb{E}[\varepsilon_t \mid d_t, x_t] = 0$ ,  $\mathbb{E}[v_t \mid x_t] = 0$ ,  $\text{Var}(v_t) > 0$ ;  $g$ ,  $m$  approximately sparse
- ▣ Post-double selection: the heteroskedasticity-robust standard error is valid under independent sampling, regularity conditions and sparsity  $s^2 \log^2(p \vee n)/n \rightarrow 0$  (Belloni, Chernozhukov & Hansen (2014)); with time series, HAC errors are an approximation
- ▣ LASSO objective  $\frac{1}{n} \sum_t (y_t - x_t^\top \beta)^2 + \frac{\lambda}{n} \sum_j \hat{\psi}_j |\beta_j|$ , plug-in  $\lambda = 2c\sqrt{n} \Phi^{-1}(1 - \gamma/(2p))$ ,  $c = 1.1$ ,  $\gamma = 0.1/\log n$ , loadings  $\hat{\psi}_j^2 = \frac{1}{n} \sum_t x_{tj}^2 \hat{e}_t^2$ , iterated
- ▣ DML (Chernozhukov et al. (2018)): with  $\ell(x) = \mathbb{E}[y \mid x]$ , it suffices (with consistency and regularity) that  $\|\hat{m} - m\| (\|\hat{\ell} - \ell\| + \|\hat{m} - m\|) = o_P(n^{-1/2})$ , e.g. both  $o_P(n^{-1/4})$

## Appendix: Machine Learning for Inference in Asset Pricing

- [Feng, Giglio & Xiu \(2020\)](#): does a new factor add pricing information given hundreds of existing ones?
  - ▶ double selection of the controlling factors from the zoo, then a valid  $t$ -test of the new factor's risk price
  - ▶ the inferential answer to [Harvey, Liu & Zhu \(2016\)](#), who raise the hurdle to  $t > 3$
- Heterogeneous effects: honest causal forests give CIs for effects that vary with covariates ([Wager & Athey \(2018\)](#))
- Three rules
  - ▶ prediction and inference are different targets: a good predictor can give a biased  $\hat{\theta}$
  - ▶ orthogonalise before you test, and cross-fit with folds that respect time
  - ▶ report CIs; “selected” or “important” is not “significant”

## Appendix: Deep Learning for Sequences

### Recurrent networks (LSTM, GRU)

- ▣ Hidden state carries memory:  $h_t = g(h_{t-1}, x_t)$
- ▣ Gates control what is kept or forgotten, mitigating vanishing gradients
- ▣ Used for volatility, limit-order-book and intraday forecasting

### Transformers

- ▣ Self-attention:  $\text{softmax}(QK^\top / \sqrt{d_k})V$ , with query, key, value matrices  $Q, K, V$
- ▣ Long-range dependence, parallel training
- ▣ Backbone of large language models ([Chapter 15](#))

## Appendix: Time-Series Foundation Models

### Pre-trained forecasters

- Pre-trained on large collections of series, used **zero-shot** (without re-training): Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), Moirai (Woo et al., 2024), Lag-Llama (Rasul et al., 2023)
  - Useful baselines for volatility and volume forecasting
  - For returns they inherit the same signal-to-noise limits
  - Pre-training data may overlap your test period: a new form of leakage
- 
- Covered in depth, with the risk-forecast evaluation tools of [Chapter 8](#), in [Chapter 14](#)

## Appendix: Reinforcement Learning, Generative Models, Explainable AI, Risk

### Reinforcement learning (RL)

- ▣ A policy  $\pi(a | s)$  chooses actions  $a$  in states  $s$  to maximise cumulative reward
- ▣ Optimal execution, market making, deep hedging: Buehler et al. (2019)

### Generative models

- ▣ GANs, variational autoencoders (VAEs), diffusion models: synthetic scenarios for stress tests

### Explainable AI

- ▣ Required by regulators (EU AI Act, credit scoring, model risk)
- ▣ SHAP, LIME (local interpretable model-agnostic explanations), partial dependence

### ML for risk

- ▣ Value-at-Risk (VaR): conditional quantiles (quantile forests, quantile-loss boosting); Expected Shortfall (ES): average of fitted tail quantiles or a joint VaR-ES loss; volatility: a squared-return target (Chapters 7–8)

## Appendix: Deep Learning in Asset Pricing, Data (Section III.A)

- Monthly returns of all CRSP stocks, January 1967 – December 2016; excess over the one-month Treasury-bill rate
- 46 firm characteristics in six groups: past returns, investment, profitability, intangibles, value, trading frictions (Table A.II)
  - ▶ Only stock-months with all 46 available: about 10,000 of 31,000 stocks; ranked each month into cross-sectional quantiles
- 178 macroeconomic series: 124 of FRED-MD ([McCracken & Ng \(2016\)](#)), the 46 monthly medians of the characteristics, 8 predictors of [Welch & Goyal \(2008\)](#); transformed to stationarity
- Split: training 1967–1986, validation 1987–1991, test 1992–2016

## Appendix: Deep Learning in Asset Pricing, Method (Sections I–II, Appendix C)

- Adversarial generalized method of moments:  $\min_{\omega} \max_g \frac{1}{N} \sum_j (E[M_{t+1} R_{t+1,j}^e g(l_t, l_{t,j})])^2$  (eq. 3);  $g$  builds the test assets with the largest pricing errors, motivated by [Hansen & Jagannathan \(1997\)](#)
- Networks: an FFN for  $\omega$ ; an LSTM that turns the macro history into 4 hidden states; the adversary  $g$  with 8 instruments; three steps: unconditional SDF, maximise over  $g$ , minimise again; the full model is labelled GAN (Section II.D)
- Tuning: 384 configurations ranked by the validation Sharpe ratio (SR); chosen: 2 layers of 64 units, dropout, Adam; ensemble of 9 fits (Table A.VIII); loadings: a second FFN fits  $E_t[R_{t+1,i}^e F_{t+1}] \propto \beta_{t,i}$
- Benchmarks: the FFN of [Gu, Kelly & Xiu \(2020\)](#); the linear SDF  $\omega_{t,i} = \theta^\top l_{t,i}$  on long and short characteristic legs, unregularised (LS) or elastic net (EN); FF3/FF5 tangency portfolios ([Fama & French \(2015\)](#))
- Metrics: SR of  $F$ ;  $EV = 1 - \hat{e}^2 / (\overline{R^e})^2$  (time averages of monthly cross-sectional means); cross-sectional  $R^2$ : one minus mean squared time-averaged residual over mean squared time-averaged excess return, stocks weighted by  $T_i/T$  (II.F)

## Appendix: Deep Learning in Asset Pricing, Results (1/2)

- Source: [Chen, Pelger & Zhu \(2024\)](#), *Management Science*
- Monthly out-of-sample SR of the SDF (Table I): GAN 0.75, EN 0.50, FFN 0.44, LS 0.42; FF5 0.22, FF3 0.19 (Table A.VI)
  - ▶ Reading: annualised 2.6 vs. 1.7 (EN) and 1.5 (FFN); the regularised linear no-arbitrage model (EN) already beats the forecasting network; unregularised LS does not
- Individual stocks (Table I): EV 8% (GAN) vs. 3–4%; cross-sectional  $R^2$  23% vs. 14–19%
  - ▶ Reading: the no-arbitrage objective improves both fits, although it maximises neither
- Deciles sorted on the loadings  $\beta$  (Table II, test period, annualised): top minus bottom 0.39; FF5 alpha 0.39,  $t = 9.96$ ; the GRS test rejects the CAPM, FF3 and FF5 ( $p = 0.00$ )
  - ▶ Reading: loadings predict returns, linearly ( $R^2 = 0.97$  of mean return on  $\beta$ , Figure 8), and the factor models do not price them

## Appendix: Deep Learning in Asset Pricing, Results (2/2)

- 46 sets of characteristic-sorted deciles (Table III): GAN cross-sectional  $R^2$  above 90% for all; EV short-term reversal 0.70 vs. 0.43 (EN), momentum 0.54 vs. 0.26
  - ▶ Reading: gains come from extreme deciles and interactions; size and book-to-market deciles are easy for every model (Section III.E)
- Macro states (Figure 6): the 178 raw series make the out-of-sample SR collapse; GAN loses about 10% without macro states and 20% with unconditional moments only
  - ▶ Reading: the dynamics of macro data must be modelled (LSTM), not fed as the last increment
- Largest stocks (Table IV, monthly SR, models fitted on all stocks, evaluated on the subset): about 1,500 stocks GAN 0.41, FFN 0.30, EN 0.15; about 550 stocks 0.26, 0.24, 0.23; re-estimated on the 1,500: 0.42, 0.28, 0.42
  - ▶ Reading: the edge narrows sharply among the largest stocks; EN catches up when re-estimated on them

## Appendix: Deep Learning in Asset Pricing, Replication and Extension

- Data limitation: the stock-level panel (CRSP returns, 46 characteristics) is not in the course data, so the GAN on individual stocks cannot be reproduced
- With the [Kenneth French Data Library](#) and FRED-MD, on the published split
  - ▶ Replicable: FF3 and FF5 tangency weights estimated on 1967–1986 (footnote 23), monthly SR on 1992–2016 next to Table A.VI
  - ▶ Not replicable with the course data: the published LS and EN (Tables I and III) use rank-weighted legs of 46 characteristics and stock-level betas
  - ▶ Analogue on portfolio data, an extension rather than a replication: the linear SDF ( $\hat{\theta}_{LS}$  in closed form,  $\hat{\theta}_{EN}$  with the objective of Section II.F) on halves (deciles 1–5, 6–10) of the 15 characteristics that the library sorts into deciles; its levels are not comparable with Tables I and III

## Appendix: Deep Learning in Asset Pricing, Limits and an Open Extension

- Limitations stated by the authors
  - ▶ The positivity of the SDF is not imposed (footnote 9); only stock-months with all characteristics, which mainly drops very small stocks (Section III.A)
  - ▶ Frictions (Section III.I, Figure 17): annual SR 1.73 without the smallest 40% of stocks, 2.07 without the widest 40% bid-ask spreads, 1.87 without the lowest 40% turnover; see also [Avramov, Cheng & Metzker \(2023\)](#)
  - ▶ In-sample metrics exceed the out-of-sample ones, a sign of some overfitting (Section III.C); the hidden states are not comparable across fits, so they cannot be averaged (Section II.E)
- Open extension: freeze the 1967–1991 design and evaluate 2017–2026, after the paper's sample; does the linear no-arbitrage SDF still beat FF5, and do LSTM states of FRED-MD add to it?

## Appendix: Deep Learning in Asset Pricing, Reading the Two Charts

- GAN 0.75 vs. FF5 0.22: with Lo SEs 0.065 and 0.058, the SE of the difference is at most 0.12 whatever their correlation, so the gap exceeds 4 SEs (i.i.d. Normal returns); a paired test on aligned returns ([Ledoit & Wolf \(2008\)](#)) is sharper
- The fixed FF5 weights load mainly on profitability and investment: RMW 0.42, CMA 0.30 (weights scaled to  $\sum |w| = 1$ ); FF3 puts 0.61 on HML; both combinations were chosen on 1967–1986
- After 2016 the fixed tangency portfolios have monthly Sharpe ratios far below the market's; with  $T = 116$  months the SE is about 0.09: a warning, not yet a test

 MFM\_ch13\_sdf\_replication

## Appendix: The Discovery Loop, Step by Step (1/2)

- **Literature:** "Summarise the evidence on how much published return predictability survives out of sample and after publication: DOI or arXiv id, sample, headline decay estimate."
  - ▶ Resolve every identifier; Semantic Scholar, Google Scholar, Elicit for the search
- **Hypotheses:** "For a feature that predicts S&P 500 returns in 2001-2015, give competing explanations (risk premium, reversal, microstructure, data mining) and what each predicts for 2016-2026 and for other indices."
- **Data and code:** "Write Python that builds 34 technical features from the course's daily S&P 500 and VIX closes with information up to day  $t$  only, and reports Newey-West  $t$ -statistics for 1-, 5- and 20-day returns in 2001-2015."
  - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot); first reproduce the case study: "always up" 58.1%

## Appendix: The Discovery Loop, Step by Step (2/2)

- **Robustness**, fixed in advance: "Write the pre-analysis plan: the full list of features and horizons, the holdout, the corrections (Holm, BH, max-t) and the result that would count as a failure."
  - ▶ Placebo: shifting the target circularly (250–3,000 days) breaks its alignment with the features but keeps each series' own dependence; a valid null for  $\max_j |t_j|$  if the target is stationary and the shifts exceed its dependence range
- **Critique**: "Act as a referee: which of these significant features are one effect counted several times, and which are known anomalies rather than new ones?"
- **Report**: all  $M$  tests, all trials including the discarded ones, DSR and PBO for any strategy built on them

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